

Overview of Google's cloud data analytics tools

As the cloud continues becoming a more prominent reach in the data world, all kinds of exciting new data tools will arise. As a data professional, it's important to know that cloud data tools will usually be specific to each stage of the data lifecycle. And there will probably be a lot of choices for each stage. So, how do you choose the right solution for a particular task? In this video, we'll introduce you to some of the tools offered by Google and explore how they can support many types of projects and initiatives.

BigQuery

First is BigQuery, a **serverless data warehouse**. This tool is pretty popular in the data world because it **works across many major cloud platforms**. With BigQuery, you use **Sequel** to query and organize data. There are also some really cool **machine learning and artificial intelligence tools** integrated within the platform. And BigQuery includes **built-in business intelligence** engines to create interactive and responsive data insights.

As a data analyst, you may use BigQuery to process seemingly unlimited amounts of data - think terabytes and even more terabytes - without having to manage the database.

Looker

Another solution that may become a part of your toolkit is Looker. Looker **organizes business data** and **builds workflows and applications** based on data insights. But it's primarily a data **visualization and reporting** tool.

As a data analyst, you might use Looker to integrate various file types like CSV, JSON, and Excel into a single application. You may also use it to publish data in a variety of dashboards.

Managed Service for Apache Spark

Next is Managed Service for Apache Spark. This service is **fully** managed and allows you to **run** Apache Hadoop, Apache Spark, and Apache Flink, along with many other open-source tools and frameworks. Managed Service for Apache Spark can **modernize** data lakes and perform ETL functions—or extract, transform, and load—for massive amounts of data.

As a data professional, you might use Managed Service for Apache Spark to process large data workloads and place them in clusters that allow you to access your data quickly.

Dataflow

Then, there's Dataflow, which gives you the ability to **stream** and batch process data in a serverless application. It's also a **serverless** app.

Data professionals may use Dataflow to develop data processing pipelines. It can assist with reading the data from its original source, transforming it, and writing it back to your database.

Cloud Data Fusion

You may also work with Cloud Data Fusion, another **fully** managed service that allows you to **integrate** multiple datasets of any size.

As a cloud data analyst, you might use Cloud Data Fusion because it allows you to use a graphical user interface instead of code to manage your data pipelines.

Knowledge Catalog

Knowledge Catalog also allows you to **work** with multiple data sources. It creates a **central** hub for **managing** and monitoring data to work across various data lakes, data warehouses, and data marts.

As a data analyst, you may access Knowledge Catalog with BigQuery. This will allow you to access data from a variety of databases and platforms with a single interface.

Lakehouse for Apache Iceberg (Lakehouse)

Finally, Lakehouse is a storage engine that you can use to **unify data warehouses and lakes** with **BigQuery and open-source frameworks**. It also provides options for **access control**, and **multi-cloud storage**.

These are just some of the standard tools you might encounter as a data analyst, and there are even more on the horizon.